

Lejun Min

Audio ML Researcher & Artist

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EDUCATION

Center for Computer Research in Music and Acoustics, Stanford University California, United States

Master of Arts in *Music, Science, and Technology*

Sept. 2024 – Expected Jan. 2027

- GPA: 4.0 / 4.0.
- Advisor: Prof. Marina Bosi, Prof. Takako Fujioka.

Zhiyuan College, Shanghai Jiao Tong University

Shanghai, China

Bachelor of Engineering in *Computer Science* (Fellowship)

Sept. 2019 – June 2023

- Member of **ACM Honor Class**, an elite CS program for **top 5%** students.
- Graduated with **top 1% Outstanding Bachelor's Thesis**.
- GPA: 89/100 (ranking: 5/27).

PUBLICATIONS

L.Min, et al., “Semantic Refinement of Universal Audio Representations through Audio-Description Alignment”, *arXiv preprint arXiv:2609.08429*, September 2026. [[arXiv](#)]

J. Dai, X. Duan, X. Fan, Y. Feng, J. Li, X. Li, Y. Li, **L. Min**, et al., “Qwen-Audio-3.0-Gen-Preview Technical Report”, *arXiv preprint arXiv:2607.27011*, July 2026. (**Equal contribution; alphabetical by family name.**) [[arXiv](#)]

L. Min, S. Chen, M. Bosi, “Leveraging Rotational M/S Coding and Machine Learning in Stereo Audio Coding”, in *International Workshop on Sound Signal Processing Applications (IWSSPA 2025)*, Costa Ballena, Spain, July 2025. [[Program](#)]

X. Qu, Y. Bai, Y. Ma, Z. Zhou, K. Lo, J. Liu, R. Yuan, **L. Min**, et al., “MuPT: A Generative Symbolic Music Pretrained Transformer”, in *Proc. 13th International Conference on Learning Representations (ICLR 2025)*, Singapore, April 2025. [[arXiv](#)] [[OpenReview](#)] [[Demo](#)]

Z. Wang, **L. Min**, G. Xia, “Whole-song Hierarchical Generation of Symbolic Music Using Cascaded Diffusion Models”, **Spotlight (top 5%)** in *Proc. 12th International Conference on Learning Representations (ICLR 2024)*, Vienna, Austria, May 2024. [[arXiv](#)] [[OpenReview](#)] [[Demo](#)]

L. Min, J. Jiang, G. Xia, J. Zhao, “Polyffusion: A Diffusion Model for Polyphonic Score Generation with Internal and External Controls”, in *Proc. 24th International Society for Music Information Retrieval Conference (ISMIR 2023)*, Milan, Italy, November 2023. [[arXiv](#)] [[Poster](#)] [[Demo](#)]

RESEARCH EXPERIENCE

Qwen Audio, Tongyi Lab, Alibaba Token Foundry, Research Intern

June 2026 – Present

- Co-developed the public Qwen-Audio-3.0-Gen-Preview model for long-form, multi-character audio scenes with temporal control.
- Improved the semantic alignment of its residual-vector-quantized audio tokenizer by **25%** through contrastive learning and multi-stage training.
- Contributed to the curation and post-processing of **1M+ hours** of audio data.
- Supervisor: Han Zhao.

Smule Labs, Research Intern

Oct. 2025 – Dec. 2025, Mar. 2026 – May 2026

- Developed an end-to-end audio-captioning framework for disentangled music-semantic representations; achieved **>0.4 Spearman** audio-caption alignment and validated specialization across **five musical attributes** in internal evaluations.
- Supervisor: Yongyi Zang.

Sony Computer Science Laboratories (CSL) Paris, Research Intern
 June 2025 – Sept. 2025

- Built a two-stage generative remixing system using mix-invariant representations and conditional generation.
- Reduced the mix-to-content variation ratio by **>90%** in internal ablations, without requiring stems at inference.
- Supervisor: Dr. Stefan Lattner.

SoundPatrol & Stanford University, Research Assistant
 Feb. 2025 – May 2025

- Trained singer-identity representations for singing-voice deepfake detection; reduced EER by **10 percentage points** and achieved **96% top-1 accuracy** on the *SingFake* dataset. The system was integrated into SoundPatrol’s infringement-detection pipeline.
- Advisor: Prof. John Thickstun, Prof. Walter De Brouwer.

Music X Lab, MBZUAI, Research Assistant
 Sept. 2023 – Feb. 2024

- Developed a four-stage cascaded diffusion framework and a hierarchical symbolic language for coherent whole-song generation up to **256 measures**.
- Significantly outperformed baselines on long-range structural coherence; published as an **ICLR 2024 Spotlight (top 5%)**.
- Advisor: Prof. Gus Xia.

Music X Lab, MBZUAI, Research Assistant
 June 2022 – Dec. 2022

- Developed Polyffusion, a piano-roll DDPM supporting five generation and control tasks through masked inpainting and cross-attention.
- Achieved the best objective results on **four of five tasks** and significant gains in perceived naturalness and musicality; published at **ISMIR 2023**.
- Advisor: Prof. Gus Xia.

SKILLS

Research Specialties	Audio & Music Generation, Audio Representation Learning, Audio Captioning, Music Information Retrieval, Audio Tokenization & Neural Codecs, Audio Deepfake Detection, Perceptual Audio Coding, Digital Signal Processing, Human-Computer Interaction
Machine Learning	Diffusion Models, Transformers, Self-Supervised Learning, Multimodal Learning, Contrastive Learning
Programming Languages	Python, C++, C, Rust, Java, Bash, Lua, Verilog, WGS�
ML Frameworks	PyTorch, TensorFlow, Hugging Face, torchaudio, librosa, Accelerate, Lightning
ML Systems	Distributed Training, Linux, Docker, SLURM, MLflow, Git
Audio & Creative Tools	JUCE, ChuckK, Reaper, Pure Data, Ambisonics, Adobe Audition, FL Studio, Audacity, Blender, Adobe Premiere, Kdenlive, Krita, LaTeX, Typst

AWARDS & HONORS

CCRMA Flagship Project Award	Feb. 2025
Chiang Chen Overseas Graduate Fellowship (one of 10 awardees in Mainland China)	Jan. 2025
CCRMA Fellowship	Sept. 2024
SJTU Outstanding Bachelor’s Thesis (one of 41 awardees out of 3873 graduates in 2023)	June 2023
Longhu Scholarship (top 5% in Zhiyuan College)	Apr. 2023
SJTU Student of Merit (one awardee in each major)	Dec. 2021
Zhiyuan Honorary Scholarship	2019 – 2023

TEACHING

Perceptual Audio Coding (Music 422), Stanford University, Teaching Assistant
 Winter 2026

- Taught a review lecture on perceptual audio codec components and modernized the coding environment through a reusable template repository.

- Lecturer: Prof. Marina Bosi.

Reinforcement Learning (CS 3316), SJTU, Teaching Assistant

Spring 2023

- Designed a final project involving single- and multi-agent learning for simulated robotic hands and legged robots.
- Lecturer: Prof. Weinan Zhang.

Design and Analysis of Algorithms (AI 2615), SJTU, Teaching Assistant

Spring 2022

- Prepared well-written lecture notes and answers for assignments.
- Lecturer: Prof. Chihao Zhang.

Principle and Practice of Computer Algorithms (CS 1952), SJTU, Teaching Assistant

Summer 2021

- Designed a comprehensive ray tracing tutorial written in the Rust language. The repository received 100+ stars on GitHub.
- Lecturer: Prof. Yong Yu.

PROGRAMMING PROJECTS

Computer Graphics

Gigantic Splight (Python)

June 2022

An interactive 3D fluids simulation based on Taichi framework.

Scotty3D (C++)

Mar. 2022

A comprehensive computer graphics project covering software rasterization, interactive mesh editing, path tracing, and animation.

Ray Tracer (Rust)

Aug. 2020

A complete ray tracing engine in Rust.

Compiler & Computer Architecture

Mx Compiler (Java)

May 2021

A compiler for a Java-like language whose optimizations outperform the reference -O1 baseline.

RISC-V CPU (Verilog)

Dec. 2020

A five-stage pipelined RV32I CPU implemented and validated on an FPGA.

Python Interpreter (C++)

Feb. 2020

A Python language interpreter.

Algorithm & Data Structure

Train Ticket System (C++)

June 2020

A collaborative full-stack project featuring a custom B+ tree storage layer and a web interface.

ART PRACTICES

Live Performance & Intermedia Art

Sound Poetry, for 2-channel audio & vocalists

Dec. 2025

A musique concrète sound poem about dreams, followed by a mandarin (grape) fugue.

Umbrella, for Ambisonics audio & video

Dec. 2025

An intermedia piece exploring the nature of self and fear.

Interplanetary Concert, for 2-channel audio, video, & live performer

Oct. 2025

Breaking the fourth wall with the interplanetary teleportation system.

A Chan Conversation, for *Ambisonics audio*, *Gametrak*, & live performer May 2025
A sonic conversation with an ancient Chan Buddhist monk. Performed at the CCRMA Open House Concert 2025.

Interface & Interactive Design

Sonic Skateboard (*Arduino* & *ChuckK*) June 2025
Turns a skateboard into a musical instrument.

Talking to A Black Hole (*ChuckK* & *ChuGL* & *WGSL*) Dec. 2024
Let the noise guide you through the event horizon of a lonely black hole.

Kandinsky Sonified (*ChuckK* & *ChuGL*) Nov. 2024
An interactive audiovisual music sequencer that creates and sonifies Kandinsky-like abstract paintings.

Fireflies (*ChuckK* & *ChuGL*) Oct. 2024
An interactive music therapy journey embodying a firefly. Essentially a sound peeking visualization.

Music & Sound Art

The Backrooms: Audio Drama, for *binaural audio* Dec. 2024
The protagonist “no-clips” into a strange space and hears things beyond his comprehension.

忆久 (Memories Last Long) June 2023
A song and a music video dedicated to the Zhiyuan College graduates of 2023.

Should Have Known Better (piano & synth cover) Feb. 2023
Piano, synth, & singing recording.

晚海 (Sunset Sea) Dec. 2021
An electronic music piece published under CEM Records.

OPEN-SOURCE CONTRIBUTIONS

canvas-downloader (*Rust*), Author & Packager
A command-line tool to download and organize all your Canvas course materials.

emailbook-rs (*Rust*), Author & Packager
A minimalistic email address book with aerc integration.

chuckfmt (*Rust*), Author & Packager
A fast code formatter for ChuckK programming language.

cli-toolbox (*Shell Scripts*), Author
A curated collection of modern command-line tools and configurations for terminal-based development.

AUR (*Shell Scripts*), Packager
A collection of maintained Arch User Repository packages with automated version tracking and updates.

plugdata (*C++*), Packager
A plugin wrapper for Pure Data, featuring a new GUI made with JUCE.

COMMUNITY SERVICE

International Computer Music Conference (ICMC) 2026, Reviewer 2026

International Computer Music Conference (ICMC) 2025, Reviewer 2025

Large Language Models for Music & Audio (LLM4MA) Workshop at ISMIR 2025, Reviewer 2025

ChuckK Development Team, CCRMA, Stanford University, Contributor 2025 – Present

Game Audio Initiative, Stanford University, Founder 2025 – Present

DSP Seminar, CCRMA, Stanford University, Speaker & Contributor 2024 – Present

Arch User Repository (AUR), Maintainer (id: aik2) 2022 – Present

LANGUAGE PROFICIENCY

Mandarin Chinese (native), English (fluent), French (beginner)

TOEFL: **112** (Reading **30**, Listening **30**, Speaking **24**, Writing **28**)

GRE: Verbal **162**, Quantitative **170**, Writing **4.0**